
Investigating Critical Gaps in the Menstrual Health Space: Using Speculative Design to Imagine the Future of Periods

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Period Futures · Publication in preparation

Abstract. Menstrual health is a public health issue encompassing diverse, holistic needs that are often insufficiently supported and not clearly defined. This research aimed to explore where menstruators perceive gaps in the systems shaping menstrual health experiences, identify the unmet needs and priorities of external stakeholders, and translate those insights into potential actionable solutions. The study applied human-centered and speculative design methodology. Eight semi-structured interviews were conducted with purposefully sampled participants. Thematic analysis identified fourteen recurring themes, which were synthesised into four primary insights across educational stigma, media representation, inclusivity and product regulation. This research identified systematic challenges shaping menstruators' everyday lives. Design opportunities were derived from these insight statements to reimagine the menstrual experience for younger generations and to provide directions for better aligning menstrual health resources with lived experience. Future work should expand participant diversity and advance co-designed solutions in partnership with a broad range of menstrual product consumers.

Keywords. menstrual health; speculative design; human-centered design; grounded theory; health equity; product regulation

1 Introduction

Menstrual health remains a substantially under-researched area of healthcare with important social, cultural and political implications [9, 30]. Limited attention to menstruation within education, research and policy frameworks has contributed to persistent gaps in education, access and equity [9]. These systemic issues manifest as inadequate access to hygiene products and insufficient menstrual health education, with tangible consequences for bodily autonomy.

Educational disparities contribute significantly. The World Health Organization reports that only 39% of schools worldwide provide menstrual health education, a gap that reinforces negative gender stereotypes and discrimination [8]. A lack of education produces misconceptions about bodily autonomy, which limits resistance to harmful social norms and sustains an uncomfortable public environment around the topic [2].

Mass media plays a dominant role in societal perception. Media has gravitated toward cleaner, simplified depictions of periods, effectively conforming menstruation to an unrealistic standard [11, 27]. That representation obscures lived realities, making it harder for menstruators to discuss their experiences openly or seek support.

Knowledge gaps are compounded by longstanding gender bias in clinical research. Women and menstruators have been underrepresented in studies, perpetuating an incomplete understanding of menstrual physiology [9, 15]. Societal taboos surround menstruation across cultural contexts, framing it as shameful and discouraging open dialogue, which creates both medical and research barriers.

There are also few regulations or standards governing sanitisation and safety of menstrual products [16]. This sustains a lack of urgency while endangering user health, reinforcing a system that normalises shame

and discomfort, and advancing a narrative in which menstrual health should remain hidden and treated as a private inconvenience rather than a public concern.

There is consequently a need to understand the future trajectory of the menstrual health space and the opportunities within it. Despite existing literature on stigma and on disproportionately burdened populations, the root causes require careful evaluation, particularly across public health, education and social media. This study uses speculative and human-centered design to investigate inadequacies and stigma in the menstrual health space, in order to understand how to address root issues and drive beneficial change.

2 Methods

2.1 Design approach

This study uses speculative design methodology, a process that challenges societal norms and explores future scenarios through critical and imaginative design [32]. Speculative design has four core phases: selection for speculation, speculative exploration, speculative transformation and speculative provocation [3]. This study used speculative provocation, creating design provocations to imagine reformative future scenarios for the menstrual experience and to simulate difficult conversations about existing stigmas and norms, which helped the team identify underlying feelings surrounding menstruation.

To ground the work, human-centered design (HCD) research methodologies were used to build provocations and co-create futures with participants. HCD proceeds through three phases: inspiration, ideation and implementation. Qualitative analysis during inspiration allows interviews and observations to be synthesised into design opportunities, which then feed brainstorming and prototyping in an iterative loop [14].

From interview data, themes were elicited using grounded theory [31]. Themes were extrapolated into key insight statements and reframed into design opportunities.

2.2 Participant selection

Eight semi-structured interviews were conducted via video conferencing. Four inclusion criteria applied: age 18 or older, fluency in English, comfort using video and audio recording, and the ability to articulate thoughts and feelings. Participants were excluded for potential conflicts of interest such as prior involvement with the Period Futures organisation. Written and signed consent was obtained at least 24 hours before each interview.

Purposeful sampling, which selects participants based on characteristics relevant to the research aims [21], was used to ensure a broad range of stakeholder perspectives. This allowed deliberate recruitment across ages, career paths, education levels and political orientations, and across varied relationships to the menstrual health space including social media activists, community leaders, educators and startup founders. Interviews covered participants' motivations regarding menstrual equity, the future changes they envision, and a review of four preliminary design provocations proposed by the team. Two to three researchers attended each interview, with at least one leading and another taking notes or guiding the provocation review.

2.3 Data synthesis

Interview transcripts, recordings and detailed notes were used to gather interview-specific data. After reviewing each interview together, the team listed eight to ten key concepts per interview with a short description for each. Key concepts were those the participant emphasised, repeated or elaborated on in detail; descriptions captured their distinctive perspective. Common underlying themes were then identified across all interviews, producing fourteen themes.

2.4 Compilation, consolidation and consensus

After sorting key takeaways into themes, descriptions were developed for each of the fourteen categories and supporting quotes extracted. The fourteen categories were then consolidated into opportunity areas with associated insight statements. Insight statements are summarising expressions derived from user research that reveal unmet needs, motivations or contextual factors, articulating why a problem exists and what matters most to users, and guiding ideation by reframing raw qualitative data into design opportunities [13]. Consensus was reached by identifying root causes and premises across the original categories through an iterative process, with all insight statements created and supported by participant quotes.

The research team then held a brainstorming workshop to derive specific design opportunities, consolidating to two or three per category. Consolidation weighted potential for sustained impact and value, favouring solutions with long-term direct benefits such as educational programmes or policy change, while recognising the importance of social and interpretive solutions such as community events or art installations.

3 Results

Eight stakeholders were interviewed, including menstrual health advocates, educators, researchers, designers and community-based leaders (Table 1). Fourteen themes were elicited, from which four insight statements and associated design opportunities were developed.

Table 1. Participant demographics and interview focus.

#	Demographic	Focus
1	Female, entrepreneur (UAE)	Reducing the gender gap in data and research so innovations address usability, accessibility and equality; advocates for global menstrual equity
2	Female, educational nonprofit	Identifies education as the primary root cause of stigma; emphasises comprehensive, accessible education for all
3	Female, feminist	Grew up in an open-minded environment but sees periods as historically shamed; women's health not prioritised in product development; advocates awareness of nutrition and exercise across the cycle
4	Female, researcher and entrepreneur	Healthcare researcher turned entrepreneur; advocates less overengineered products and greater understanding of natural processes; envisions holistic products supporting long-term needs
5	Female, product designer	Emphasises catering to those with chronic reproductive disorders, diabetes and menopause; stresses economic and political context in combating stigma
6	Female, public health	Focuses on facilitating conversations that make people aware of their opportunities within the field as it currently exists politically
7	Female, sociologist	Policy reform, particularly product accessibility across workplaces and more consistent, in-depth information across education systems
8	Female, rural activist	Improving access to period products in rural areas; uplifting the voices of differently abled people

3.1 Insight 1: education

Students and educators are ready to discuss menstrual health and experiences, but deep-rooted stigma and outdated, narrow and vague policies keep menstrual education stagnant.

Menstrual education has remained largely unregulated. Only about 25.5 % of US K–12 public school education standards include instruction on menstruation, and comprehensive menstrual education covering hygiene, abnormal bleeding and stigma is increasingly rare [5]. No US state or territory currently requires that school-based menstrual education be evidence-based [12]. Policy analyses show most states lack mandated funding or accountability measures for evidence-based menstrual health education [19]. Teachers report feeling underprepared and unsupported. Federal accountability standards such as the Every Student Succeeds Act do not include menstrual health as a required topic, and schools often treat it as optional [1, 6].

Current curricula are gendered and frequently exclude children who do not menstruate. Participant 4 stated that menstrual education should be introduced in school like sex education and should include both boys and girls in middle school. For menstruators, the curriculum lacks specificity: Participant 2 described so much missing from sex education that one begins to feel something is wrong with one's own body, and was embarrassed to talk about it. Participant 3 recalled a mother handing over a pamphlet in junior high, which communicated discomfort rather than information.

Reform efforts show appetite for change. Interviews of US school communities note that menstrual health education always gets pushed aside [26], and researchers found that both adult women and female high school students are largely self-taught in menstrual hygiene despite wanting to learn more [14]. Of fifty states, only California, Michigan and New Jersey cover menstrual hygiene products, and only Michigan, Oregon and Utah cover menstruation management [14]. Even where state requirements exist there is often a disconnect between mandate and classroom practice, since district policies may compete with state ones [4]. The National Sexuality Education Standards, commonly used to develop state standards, address neither personal hygiene during menstruation nor management strategies [20].

Design opportunity 1

Reimagine menstrual health education by shifting from vague, gender-specific instruction to inclusive, evidence-based and encouraging learning experiences, normalising menstruation and equipping students with an unbiased mindset.

Education should be provided early regardless of gender and cover anatomy, hygiene, cycle specifics and management. Washington D.C. public schools demonstrate the approach, mandating menstrual health education standards for all students regardless of gender. Students should also engage in hands-on activities: exploring products, guest speakers, or a menstrual hygiene lab testing product absorbency, simulating vaginal pH or examining bacterial growth in culture samples. Re-educating educators equips them to address stigma in an evidence-based manner; Zambia's Water, Sanitation and Hygiene programme trained 55 teachers from 36 schools, covering menstruation management, the broader impact of puberty and a sanitation practice model. Educators should be provided an adaptable curriculum, teaching tools and props, and funding to tailor delivery.

3.2 Insight 2: media representation

Although conversation about menstruation has expanded over the last decade, periods have been sanitised by the media, and true openness to the reality of menstruation is not represented in many cultures.

Participant 1 described being sidelined on social media and excluded from farmers' market vendor directories after releasing a period product, on the grounds that it contained adult content. Unique innovations are being shunned for displaying the realities of menstruation, revealing a clear preference for simple, clean

depictions that do not represent lived experience and creating a disconnect between how periods are discussed and how they are.

Participants expressed a desire for more honest depiction. Participant 4 observed that prime-time sanitary product advertising slowly breaks down taboo because it removes secrecy. Participant 5 noted the increasing visibility of tampons in women's bathrooms as a positive development. There is a link between femininity and menstruation in which periods are sanitised to remain comfortable for viewers; in reality menstruation can be emotionally difficult, and being related to femininity does not require it to be clean and easy to digest.

Design opportunity 2

Shift public exposure to menstruation through intentional representation in media and entertainment, to encourage open-minded community practice and normalise conversation.

Short films and streaming advertisements that realistically portray menstruation, including mood changes, cramps and leaks within everyday storylines, can normalise menstruation, reduce shame and educate audiences about products and resources, particularly when amplified by content creators. Partnerships with emerging filmmakers, health-focused associations and product companies could make this tangible. Reframing menstruation as a shared cultural experience through art and history showcases can transform it from a private burden into a community conversation; such an exhibition would display innovations, cultural taboos, artworks and personal experiences across schools, workplaces, community centres and travelling pop-ups. An interactive website could foster an online community oriented around education and social connection, integrating emotional, cultural and social dimensions, with privacy addressed through data protection.

3.3 Insight 3: inclusivity

Menstruation has been confined to a homogeneous experience, failing to account for diverse needs, identities and health statuses.

Store shelves appear filled with modern products that in reality differ little, an illusion of progress that turns menstruation into a monolith and manufactures a narrow idea of an average menstruator. Participant 2 described being overwhelmed by choice, confronting a twenty-foot shelf of pads and tampons without knowing what was needed. Participant 5 stressed that many face chronic conditions, disabilities and hormonal disorders that shape cycle management, and that most available products focus only on the healthy body.

Consumer preferences are underexamined. Participant 4 wanted eco-friendly options that perform as well as unsustainable versions, having found that they do not hold up. Participant 2 raised chemical exposure across water, air and clothing, and asked where it could be eliminated. Translating this feedback into innovation would address menstrual care needs more effectively.

Design opportunity 3

Reframe menstrual care as a personal, adaptive practice catering to diverse life experiences.

A customisable menstrual care platform would let individuals build a personal profile capturing flow patterns, comfort preferences and health considerations, alongside preferences for gender accommodation, financial constraints, sustainability or quality priorities, and accessibility needs, then generate individualised product recommendations or care kits. Critically, the platform would grow with the user: preferences updated each cycle allow the system to adapt across puberty, pregnancy and menopause, with embedded educational content explaining why needs shift and what options exist. Partnerships with public health organisations, community clinics, social enterprises and media companies would extend access for those facing barriers.

3.4 Insight 4: regulation

The lack of standardised regulation, insufficient testing and commercial incentives compromise the quality of menstrual products and endanger health.

Medical advances have not reached the regulation and testing of menstrual products. The FDA has not substantially updated menstrual pad regulations since 2005 [17, 22, 25] even though menstrual underwear, silicone cups and absorbent discs have all entered the market since [7, 18]. Standards for testing across diverse subgroups are lacking, which is particularly concerning for those sensitive or allergic to compounds in products. A 2024 study from the University of California, Berkeley found detectable levels of toxic metals including lead and arsenic in several menstrual products [10, 28]. In 2023, investigators identified per- and polyfluoroalkyl substances (PFAS) in products marketed as PFAS-free [24, 29]. The Environmental Working Group has reported undisclosed chemicals, potential endocrine disruptors and allergens in many period products [23].

Current pre-market testing covers basic absorbency, chemical, physical strength and toxicological measures. There is no requirement to assess long-term risks from chemical and plastic exposure, so chronic impacts remain unclear. Only three states, California, New York and Minnesota, currently require full ingredient disclosure [33]. Scholars describe this concealment as the menstrual concealment imperative, in which social norms enforce hiddenness and allow companies to prioritise profit over transparency [22].

Design opportunity 4

Redefine the standards governing menstrual products to enable greater transparency, stricter adherence and inclusivity.

What would it mean to redesign the rules from the ground up, and to treat menstrual products as long-term exposure devices rather than disposable commodities? Instead of basic absorbency and short-term toxicology, testing would assess long-term exposure risk and safety across diverse populations, including those with different hormone profiles, chemical sensitivities, health conditions and accessibility needs. Full ingredient disclosure and clear allergen guidance would address transparency, and consistent labelling would allow informed choice given growing research on contaminants.

Addressing the data gap at its root means longitudinal cohort studies following menstruators for five to ten years, with semiannual check-ins examining hormonal disruption, fertility changes, autoimmune flares, reproductive disorders and cancer risk, all of which can have latency or cumulative-dose relationships that short studies miss. These studies should include the populations named above as well as trans and non-binary individuals. The goal is not only to prove products safe, but to understand for whom, under what conditions and with what consequences.

4 Discussion

Speculative design combined with the human-centered design framework allowed common themes to be identified and insight statements compiled, and provided a flexible guide for conducting and synthesising qualitative data where diversity of perspective is integral. The volume of insight allowed the team to map distinct flows of thought, classify findings systematically and translate them into actionable solutions.

The findings support a correlation between pre-existing conceptions of menstruation and dissatisfaction among current menstruators. Those conceptions are largely shaped by generational disparities in understanding, which further analysis dissected into gaps in education, experience, media, regulation and testing.

On education, participants expressed an unmet need for instruction that is comprehensible and adaptive, and emphasised that current approaches have remained static despite changes in the social and informational

landscape. As a widely experienced public health issue, menstruation warrants comprehensive integration into educational systems, and addressing deficiencies in content and delivery is critical to reducing misinformation.

On media, participants voiced that limited representation sustains shame. Portrayals remain entrenched in generational taboos, and understanding how those views enter media representation is necessary before it can be changed.

The homogeneous menstrual experience was discussed frequently, with interviewees expressing a need for greater diversity to account for different period types and experiences. This aligns directly with the final unmet need: insufficient regulation and testing. Participants sought increased transparency, tighter regulation and more extensive testing, noting that many mainstream products fail to meet their stated claims. Inadequate oversight combined with outdated standards reduces product quality and user satisfaction.

These unmet needs point to a broader pattern of generational stagnation affecting both user experience and the narratives through which menstruation is understood. Interview findings revealed that generational gaps strain communication, reduce comfort and undermine self-confidence.

The findings align with existing literature, particularly on structural and cultural stigma. A Spain-based study on menstrual social dynamics argues that stigma operates not only interpersonally but through structural and cultural mechanisms including media representation, education systems and normative gender scripts [27], a pattern reflected directly in these interviews. Research conducted in Italy suggests willingness to reform menstrual education alongside institutional barriers that hinder change, with participants describing the education they received as minimal or insufficient, matching the pattern found here. The literature further supports the finding that menstruation has been framed as a homogeneous, non-individual experience, with discussion historically confined to mothers and daughters and male family members excluded, reinforcing the idea that menstruation is a private female issue rather than a health concern. Generational differences complicate this: older menstruators often perceive greater normalisation relative to the past, while younger generations note persistent taboos and inequalities.

5 Limitations

Despite diversity among participants in geography, culture, age, socioeconomic status and occupation, findings may be limited in generalisability by the small sample size. Some design opportunities apply only to populations with access to the resources and technology required to use or benefit from them.

During analysis the team encountered contradictory statements. Most notably, several participants commented on the lack of diverse products able to accommodate concurrent conditions, while another described a tyranny of choice in a market flooded with options. Reconciling these remains open, and raises further questions: how does someone's background shape their perspective on product availability, and are some areas of women's health more abundant in genuinely differentiated products than others?

6 Conclusion

The primary goal of this paper was to examine how social stigma and structural inequities across education, media, research and product regulation shape modern menstrual health experiences. The speculative design approach guided conversations with health equity stakeholders, allowing recurring themes to be identified and translated into potential solutions.

Addressing gaps in the menstrual health space requires coordinated action. Educators, researchers, designers and policymakers must move beyond symbolic inclusion and commit to evidence-based curricula, transparent regulation and inclusive product development. Media creators and community leaders play critical roles in challenging stigma through accurate representation. By aligning institutional practice with lived experience, the menstrual health space can shift from one defined by silence and inequity to one grounded in education and informed choice.

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Document note. Citations were renumbered into a single sequence and normalised to a consistent style; several entries in the source reference list were incomplete (missing volume, page or publisher details) and have been left as complete as the source allowed rather than reconstructed. Supplementary Table 1, the fourteen-category theme table with supporting quotations, is retained in the original manuscript and not reproduced here. Figure 1, the speculative design methodology diagram, is likewise not reproduced.